

Data Science Project · UGST4092

Quantifying Anthropogenic Influence on Spatial Avian Distribution

Group B

Scientific Report

Group members

Mia Corduneanu-Huci

Shiwam K.C.

Thanika Haltrich

Course instructors: Jascha Gröbel, Bojan Evkoski

Submission date: April 12, 2026

1 Introduction

1.1 Motivation and Relevance

Human expansion has fundamentally altered the natural environment. One consequence that is caused by human activity is the disruption of nocturnal ecosystems through artificial light at night (ALAN). Satellite-based analyses indicate that over 80% of the world's population now lives under artificially lit skies, with the lit surface of Earth expanding at an estimated rate of 2.2% per year between 2012 and 2016 [1]. More recent observations suggest that night sky brightness is increasing by 7–10% annually in parts of North America and Europe. Beyond light pollution, human activity generates air and particulate pollution, drives urbanisation, and reshapes land use patterns. All of these aforementioned alterations have an impact on wildlife.

Birds, particularly migratory species, serve as sensitive ecological indicators of environmental health. Approximately 80% of North American migratory bird species travel primarily at night, relying on celestial cues, geomagnetic orientation, and circadian rhythms for navigation [2, 3]. Artificial light disrupts these mechanisms, attracting birds toward illuminated urban centres where they face exhaustion, disorientation, predation, and collisions with glass-fronted buildings. Long-term monitoring programs, such as those at Chicago's McCormick Place convention center, have documented over 40,000 collision fatalities over four decades, and building collisions linked to ALAN are estimated to cause hundreds of millions of bird deaths annually across North America [4, 5].

Despite these established links, quantitative analyses of *how much* and *in what ways* different dimensions of human activity collectively influence avian migration trajectories remain limited. Existing forecasting platforms, such as BirdCast, use atmospheric and radar data to predict migration intensity but do not incorporate factors caused by human expansion such as nightlight intensity or air quality indicators. This project addressed this gap directly.

1.2 Research Objectives and Scope

This project aims to **quantify the influence of anthropogenic activity on avian behaviour** in Europe between 2017 and 2022. We focus on two primary proxies for human activity: (1) nighttime light intensity derived from NASA satellite imagery, and (2) air quality indicators. Furthermore, we have chosen to conduct our research in 2017 to 2022, a time that encompassed the COVID-19 pandemic. This is because the lockdown caused a sharp decline in human activity, energy consumption, and pollution. This allows us to do quasi-experimental assessment of how reduced human activity affected bird spatial behaviour.

Our central hypothesis is:

H_1 : Human activity has a significant effect on the spatial distribution of birds.

To test the hypothesis, we broke down the test into two sub-hypotheses: ($H_{1.1}$) nightlights have a significant effect on the spatial distribution of birds, and ($H_{1.2}$) air pollution has a significant effect on the spatial distribution of birds. We test these hypotheses by analysing whether birds are observed systematically closer to or farther from urban centres in response to changes in nightlight intensity and pollution levels.

The results are intended to benefit conservation organisations, urban planners, policy makers,

and researchers seeking data-driven insights for biodiversity monitoring and wildlife-friendly development strategies.

2 Background: Data and Problem

2.1 Background

Understanding how anthropogenic activity shapes the spatial distribution of wildlife requires both appropriate proxies for human pressure and analytical frameworks capable of isolating human effects from ecological confounders. Two recent studies provide examples of different methodological precedents for the this project, each addressing a different but complementary dimension of this problem.

Sun et al. (2022) conducted the first continental-scale analysis of urbanisation’s effects on bird distribution in China, combining citizen science occurrence records from eBird, GBIF, and the China Bird Watching Record Center with satellite-derived measures of human activity [15]. Their primary proxies for urbanisation were nighttime light intensity, drawn from the NPP-VIIRS satellite product at 500-metre spatial resolution, and impervious surface area from the Global Annual Impervious Area dataset — both tracked annually from 2000 to 2018. They built Species Distribution Models (SDMs) for 1,021 species using five modelling algorithms and 19 bioclimatic variables, producing spatially explicit predictions of where each species could potentially occur under climate alone. They then computed a the ratio of a species’ observed to predicted distribution area, to quantify how much of its suitable habitat a species actually occupies, with shortfalls attributed to anthropogenic pressure. Their results showed that birds preferring wetland and forest habitats had significantly lower range filling than urban-adapted generalists, and that hotspots of threatened bird distribution overlapped substantially with areas of the fastest-growing nighttime light intensity. Beyond individual species, they did also a between-city bird network. This study establishes that satellite-derived nighttime light data is a valid and operational proxy for quantifying the influence of human activity on bird spatial distributions at large scales, the same methodological principle as this project.

While Sun et al. demonstrate the association between nighttime lights and bird distribution at a macro scale, Kavwele et al. (2022) provide a complementary framework for isolating the causal direction of human influence through a quasi-experimental design. The study combined data collection, machine learning, and statistical modelling to analyse wildlife spatial distribution. Data were collected using a network of camera traps positioned along transects extending from the boundary of Serengeti National Park into its interior. The purpose was to observe animals relative to distance from human–ecosystem edges [16]. The large volume of image data was processed using a deep learning object detection model (YOLO) and manual validation to ensure accuracy. Environmental covariates, including vegetation dynamics (NDVI and its temporal change) and grass nutrient content, were integrated from remote sensing and spatial interpolation methods. The analysis applies a Bayesian framework, specifically a generalized linear model with a Bernoulli likelihood, to estimate the probability of species presence as a function of distance to boundary and environmental factors. Model parameters were inferred using Markov Chain Monte Carlo (MCMC) techniques, allowing for probabilistic interpretation of results and assessment of uncertainty. This study demonstrates that human activity non-locally reshapes migratory animal spatial distributions and this motivates our hypothesis that European birds

are systematically displaced closer to or farther from urban centers in response to variation in nighttime light intensity and air pollution.

These two studies establish two foundational concepts for our analysis. First, satellite-derived nighttime light intensity is a reliable proxy for anthropogenic pressure and second, variation in human activity levels (whether from land-use type or, in our case, the COVID-19 pandemic) can function as a quasi-experimental treatment allowing causal inference about how reduced human pressure affects wildlife spatial behaviour.

2.2 The Datasets

This project integrates five distinct data sources, each capturing a different dimension of the human–bird interaction. The study area is Europe (latitude 35°N–72°N, longitude 25°W–65°E), and the temporal scope is from the year 2017 to 2022, with specific attention to the COVID-19 disruption year of 2020.

2.2.1 eBird (Cornell Lab of Ornithology)

eBird is a citizen-science platform through which millions of birdwatchers worldwide submit observation records. For this project, data was obtained by directly requesting species-specific data. We chose to obtain 18 bird species: 12 migratory and 6 native species. Native species serve as a control group to isolate migration-specific responses from general avian behavioural trends. The migratory species are further subdivided by seasonal migration type (fall-only migrants vs. spring-and-fall migrants) and activity pattern (diurnal vs. nocturnal). After data collection, the raw dataset comprised 4,024,859 observations for migratory species (totalling 10,763,547 individual birds counted) and 873,931 observations for native species (totalling 15,795,542 birds counted).

A critical bias inherent to eBird is the *human footprint bias*: observations are clustered where birders go, not necessarily where birds are most abundant. A second bias is the *missing zeros* problem, in which absence of a sighting is ambiguous between genuine absence and non-reporting. This issue is further elaborated in the sections 2.2 and 5.2. However, this dataset still offers insight into spatial bird distributions throughout the years. The results we have obtained using this dataset were useful, albeit subject to bias.

2.2.2 NASA VIIRS Day/Night Band Nighttime Lights (VNP46A4)

This satellite images dataset has daily, pixel-level nighttime radiance values at a spatial resolution, which is good for urban analysis. For Europe, the dataset includes 22 tiles, each of 2400×2400 pixels, resulting in 132 files over the six years study period that the project focuses on. Key quality attributes are variability brightness values, snow-bias correction, all-angle observations, land/water pixel classification, and other "layers" which are used for data cleaning. But, due to the data complexity, we opted to use the yearly aggregated brightness. We decided to use this approach since it still captures the effect of nightlight brightness throughout the years, and it is computational easier to work with it. With this dataset, we extracted the average brightness of cities throughout the years to see whether or not nightlight intensity had an effect on the spatial distribution of birds.

2.2.3 Air Quality Index (AQI)

Real-time and historical air quality data were obtained from the AQI platform, which aggregates readings from monitoring stations worldwide at the city and town level. One of the pollutant variables is PM10. PM10 (particulate matter $\leq 10 \mu\text{m}$) values for European cities between 2017 and 2022 were used. The pollution dataset covered 241 European cities, substantially fewer than the roughly 830 cities represented in the eBird data. To bridge this gap, bird observations were spatially matched to the nearest city with available AQI data, with a 25 km distance threshold applied to ensure ecological relevance.

2.2.4 World Bank Development Indicators (WDI)

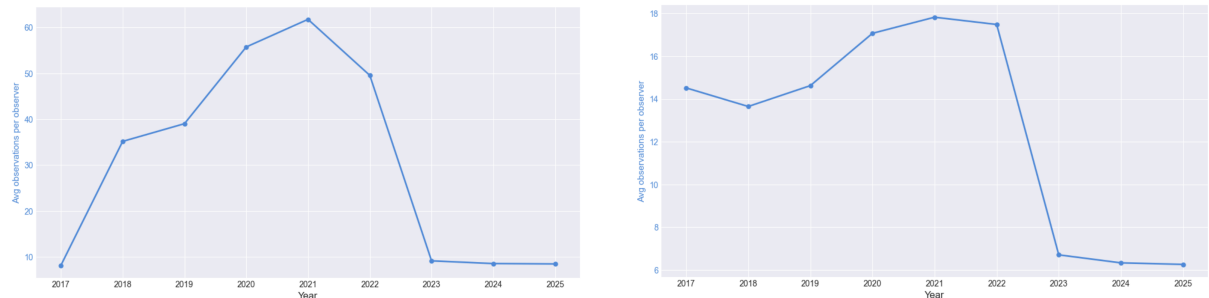
Country-level socioeconomic and environmental indicators from the World Bank DataBank were also used to capture the effect of human expansion from a different viewpoint. The dataset covers 44 European countries across 8 years (352 observations) and includes 70 features. In the end, we selected the GDP data per capita as the scalable proxy for human activity density and urbanisation.

2.2.5 Natural Earth populated places dataset

European city locations were sourced from the Natural Earth 1:10m Populated Places dataset, a high-resolution public dataset of global settlements. European area limits were applied in order to filter for the project focus area (longitudes -25° to 45° , latitudes 34° to 72°) and resulted in 1,148 cities, providing the geographic reference points used to compute bird observation distances to urban centers.

2.3 Problem Framing and Complexity

An analytical challenge we first realized in this project is distinguishing *genuine* changes in bird spatial behaviour from biases caused during the data collection process. This problem is caused since the eBird platform relies on humans manually making observations. This means that observer activity varies by year, location, and species popularity. Notably, 2022 was eBird's 20th anniversary, which caused a spike in observer participation and inflated observation counts. We have acknowledged these confounders. However, due to the scope of the course, we decided to continue performing statistical inferences using this dataset.



(a) average number of native bird observations per e-birder

(b) average number of migratory bird observations per e-birder

Figure 1: The average number of bird observations per e-birder. These figures illustrate how the ebird platform was highly active until the year 2023, when it became significantly less active.

A further complexity is the spatial mismatch between datasets: nightlight data are pixel-based raster files, AQI data are station-based point data, and bird sightings are coordinate-based occurrences. Integrating these requires spatial joins, coordinate-to-tile mapping, and proximity-based matching. The different effects COVID-19 had on nightlight intensity across cities (some cities brightened, others dimmed) further complicates the causal interpretation of lockdown-era changes and made statistical tests for this period harder to conduct.

3 Repository: project work

GitHub: Anthropogenic Influence on Spatial Avian Distribution Repository

Table 1: Overview of repository branches.

Branch	Description
<code>final_version</code>	Clean final notebooks and report. Tagged as v1.0 (this is the official submission)
<code>main</code>	Merged version containing the final files

Branches `all-work`, `mia_nightlights`, `nika_nightlights`, `nika_merged_data` include the work done for the project, original exploratory work and intermediate analyses.

To reproduce the analysis, follow the next steps: clone the repository, switch to the `final_version` branch and install all required dependencies(`pip install -r requirements.txt`). Download the raw data from the sources listed in Section 2.2 and place the files in a local `/data/` folder. Note that due to file size constraints, data and other files are stored in the an OneDrive provided in the `night_lights_notebook`). In order to run the notebooks in `/notebooks/`, note that the data collection and individual analysis notebooks must be run before the merged dataset analysis. Once each notebook has been run at least once, its output is automatically saved as a `.csv` file to the shared `/data/` directory, so that all next notebooks load the correct preprocessed datasets without needing to re-run earlier steps.

The final version of the project is tagged as **v1.0** on the `final_version` branch and the project is licensed under the **Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0)** license.

4 Methodology

4.1 Overview of the Pipeline

Data Collection → data cleaning and preprocessing → exploratory analysis → feature engineering and spatial integration → hypothesis testing → interpret results and make conclusions. Please check appendix A for the complete flowchart.

4.2 Data Collection

Data were collected from four sources. **eBird** observation records were obtained initially via eBird API, then we switched to direct bulk requests since it was simpler. The 18 target species (12 migratory and 6 native) were selected based on which period they are seen in Europe, seasonal migration period (fall only migrants and fall-spring migrants), and activity pattern (nocturnal vs. diurnal). **NASA VIIRS VNP46A4** nighttime radiance files were downloaded in HDF5 format from the NASA EOSDIS Land Processes DAAC; 22 tiles were required to cover the European study area (35°N–72°N, 25°W–65°E) across 2017–2022. **AQI** provided city-level air quality for approximately 240 European cities. **World Bank Development Indicators (WDI)** supplied country-level GDP per capita and agricultural land change data for 44 European countries across 8 years (352 observations, 70 features).

4.3 Data Cleaning & Preprocessing

eBird: Records were filtered to the European bounding box and the 2017–2025 period. We filtered only European observations. Column selection retained only variables relevant to the analysis. Outlier removal addressed extreme count values (up to 140,000 individuals in a single migratory observation and 10,000 for native species) by defining a specific upper bound. Afterwards, we analyzed the number of observations per year of each species. We realized that in the years 2023 - 2025 there were no bird sightings of some species, which could ruin the results of our analysis, so we discarded the observations during this time period altogether.

NASA VIIRS: A custom extraction pipeline mapped each European city’s coordinates to its corresponding VIIRS tile and pixel, extracting annual average radiance. Year-over-year differences (e.g. 2022 - 2017) and COVID-specific changes (2019 to 2020) were computed and individual analysis was done. For the data cleaning, the images are constructed based on "layers" that show different aspects of the image. In order to make the image clear and the nightlights intensity visible the "layers" need to be changed. For example, if there is snow, the nightlights might seem more intense than usual since the light will reflect in the snow. In this case the layer `HDFEOS/GRIDS/VIIRS_Grid_DNB_2d/Data Fields/AllAngle_Composite_Snow_Free` needs to have different values.

AQI & WDI: Standard outlier removal and feature selection were applied. Bird observations were spatially matched to the nearest AQI-monitored city within a 25 km threshold via a proximity join, beyond which air quality measurements were deemed ecologically irrelevant.

4.4 Exploratory Data Analysis (EDA)

EDA was conducted across all four (no EDA for the cities dataset) datasets to understand overall distributions, temporal trends, and potential confounders. Key findings that affected the later stages of this project include: a trend of increasing eBird observations that peaked in 2022, then a sharp decline in the number of observations in 2023. This led to extreme results in our analysis. In the end we dropped observations in the year 2023 - 2025; high heterogeneity in COVID-era NTL changes across cities; and the top 10 brightest European cities by average radiance (led by Zaragoza at 102.1, Athens at 101.1, and London at 99.5 units). An interactive map was produced using `folium`, overlaying bird observation density with nightlight radiance to visually assess spatial co-occurrence. PCA was applied to the 70-feature WDI matrix to identify

the main axes of variation across European countries. The other distributions, temporal trends of eBird results can be found in the appendix C. The `html` file of the interactive map can be found in our Github repository.

4.5 Feature Engineering & Spatial Integration

The central derived feature is the **distance from each bird observation to the nearest city centre** (km). This variable operationalises the core research question: are birds observed closer to or farther from urban centres in response to changes in nightlight intensity and pollution? Annual mean and median distances were computed per species subgroup and country to serve as inputs for hypothesis testing. All four datasets were joined at this stage: VIIRS radiance values and AQI results were matched to each observation via city coordinates, and WDI indicators were merged at the country-year level. The result was linking each bird observation to its distance from a city, the nightlight intensity of that city, its air quality, and the broader development context of its country.

4.6 Hypothesis Testing

The hypotheses tested are stated in Table 2. The hypotheses $H_{1.3}$ and $H_{1.4}$ were defined as the natural extension for us to better claim our main hypothesis H_1 . $H_{1.3}$ was tested to see if birds significantly got closer to cities or not; and to compare if one bird subgroup got more closer to a city relative to another. $H_{1.4}$ was tested to see whether the COVID-19 lockdown had an affect on spatial bird distribution. To test all of these hypotheses, three main statistical approaches were used:

Weighted Least Squares (WLS) Regression was fitted for each bird subgroup (native vs. migratory; fall vs. spring migrants; diurnal vs. nocturnal) with year as the predictor and mean distance-to-city as the response. Observations were weighted by stratum sample size so that year-country cells with more records contributed more to the estimate. This yielded slope estimates (km/year) and associated p-values for urban proximity trends.

Difference-in-Differences (DiD) exploited the COVID-19 lockdowns as a natural experiment. DiD was applied for 2 approaches. First the focus was on high vs low NTL cities around COVID. Cities above the median nightlight intensity in 2019 are the treatment group, cities below are the control. The "treatment" is the COVID period (2020–2021), since human activity and implicit also nightlights dropped in many cities. The second DiD design focuses migratory vs native species around COVID. Migratory species are the treatment group (hypothesised to be more sensitive to light and human disturbance); native/resident species are the control. The treatment period is COVID (2020–2021).

4.7 Simple and Multivariate Linear Regression

To examine whether the nightlight effect differs between species types, the NTL and $\log(\text{distance})$ relationship was estimated separately for native and migratory birds at the city-year aggregated level, with a regression line fitted for each group. This allows us to test whether permanent residents (exposed to artificial light year-round) respond differently to urban light intensity than migratory species that pass through cities temporarily.

This model extends the nightlights-only specification by incorporating air quality data. Bird observations from the native species dataset (with nightlight intensity already merged) are joined with city-level annual pollution scores from AQI. The dependent variable is `distance_km` (raw kilometres to the nearest city). The feature set consists of four predictors: `ntl_intensity`, `Year`, `month` and `pollution_score_weighted`, a composite air quality index compiled, weighting individual pollutants (CO, NO₂, SO₂, PM₁₀, PM_{2.5}, O₃) by their relative health impact.

Table 2: Overview of all hypotheses tested in this project.

ID	Hypothesis
<i>Main Hypothesis</i>	
H ₁	Human activity has a significant effect on the spatial distribution of birds.
<i>Sub-Hypotheses</i>	
H _{1.1}	Nightlight intensity has a significant effect on the spatial distribution of birds.
H _{1.2}	Air pollution (PM10) has a significant effect on the spatial distribution of birds.
<i>Subgroup Comparisons (WLS and Mann–Kendall)</i>	
H _{1.3a}	Native birds show a significant trend in distance-to-city over the years.
H _{1.3b}	Migratory birds show a significant trend in distance-to-city over the years.
H _{1.3c}	The trend in distance-to-city differs significantly between native and migratory birds.
H _{1.3d}	Fall migrating birds show a significant trend in distance-to-city over the years.
H _{1.3e}	Spring-and-fall migrating birds show a significant trend in distance-to-city over the years.
H _{1.3f}	The trend differs significantly between fall and spring-and-fall migrating birds.
H _{1.3g}	Diurnal migrating birds show a significant trend in distance-to-city over the years.
H _{1.3h}	Nocturnal migrating birds show a significant trend in distance-to-city over the years.
H _{1.3i}	The trend differs significantly between diurnal and nocturnal migrating birds.
<i>COVID-19 Natural Experiment (DiD)</i>	
H _{1.4a}	The reduction in nightlight intensity during COVID-19 lockdowns caused a significant change in bird distance-to-city.
H _{1.4b}	The COVID-19 effect on bird distance-to-city differed significantly between native and migratory species.
<i>Simple & Multivariate Linear Regression</i>	
H _{1.5a}	Nightlights influence the birds' distance-to-city over the years.
H _{1.5b}	The COVID-19 effect on bird distance-to-city differed significantly between native and migratory species.

4.8 Robustness Checks

To verify the reliability of the WLS results for the subgroup trend hypotheses ($H_{1.3a}$ – $H_{1.3i}$), two robustness checks were performed:

Mann–Kendall Trend Test: the non-parametric Mann–Kendall test was applied to the annual distance-to-city series for each bird subgroup. Unlike WLS, this test makes no distributional assumptions and detects monotonic trends without being sensitive to the magnitude of individual observations. Due to the short time series (6 years), an exact permutation variant was used to ensure more reliable p-values.

Median-based Analysis: both the WLS regression and the Mann–Kendall test were additionally run using annual *median* distances rather than means. Medians are less sensitive to outliers than means, so comparing the two sets of results allowed us to assess whether the findings were driven by a small number of extreme observations or reflected a genuine central tendency in each bird subgroup.

Result interpretation: we interpreted a result as statistically significant only when all four tests had the same outcome. Other results were deemed inconclusive.

5 Results Summary

5.1 Bird Distance to Cities: Subgroup Trend Results

Summarized results can be found in **appendix B**, Table 3 for hypotheses $H_{1.3a}$ - $H_{1.3i}$. The WLS regression and Mann-Kendall trend test using both means and medians were used to test all the hypotheses. The graphs illustrating the trends of birds distance to cities for all bird subgroups can be found in the **appendix D**.

Across all subgroups, mean-based WLS consistently returned significant negative distance-to-city trends, indicating birds get closer to urban centres over time. However, these results largely failed the median-based robustness checks, suggesting the effect is driven by a shifting subset of birds rather than a broad population-level trend. The one finding that passed all robustness checks was the diurnal vs. nocturnal comparison ($H_{1.3i}$). The difference of the WLS slopes between the two groups was +4.2783 km/year ($SE = 0.7310$, $t = 5.853$, $p = 0.0004$), indicating that the distance to cities of the two groups diverged significantly over time. All other subgroup comparisons were deemed inconclusive.

5.2 Robustness Checks

Mann–Kendall tests applied to median distances returned mostly non-significant results, whereas WLS on mean distances yielded significance. This discrepancy is interpreted as follows: a majority of birds showed stable city distances over time, but a subset showed a meaningful shift toward urban areas. Means capture this shifting subset while medians reflect the stable majority. This interpretation motivates the recommendation to extend species coverage in future work.

5.3 World Bank Indicators

We used World Bank Development Indicators dataset to do country-level analysis. For example, we wanted to check if agricultural land change across European countries correlates with migratory

bird species diversity. We computed both Pearson ($r = -0.089$, $p = 0.1452$) and Spearman ($r = -0.199$, $p = 0.0010$) correlations between agricultural land share and species-per-checklist. The Spearman result is telling: a significant monotonic relationship exists, even where the linear one does not. We then fitted per-country linear trends for both agricultural land coverage and species diversity, finding that 8 of the 27 countries analysed showed simultaneous agricultural expansion and declining species richness. We also tested countries experiencing both agricultural expansion and forest loss at the same time resulting in lower observed species.

5.4 COVID-19 Difference-in-Differences

For the first approach ($H_{1.4a}$) what stopped our analysis was the violated parallel assumption. Before COVID, the two groups were already moving in opposite directions (2017 $\rightarrow$ 2019), which does not match the DiD assumption. The groups were fundamentally different to begin with, not just in NTL level, but in how their bird observation distances were evolving. DiD = -0.030 and $p = 0.376$ so not significant. Also, for the second design DiD ($H_{1.4b}$), the parallel trends assumption were partially violated, since before COVID, native showed a declining trend in distance while migrants were stable. This limits strict causal interpretation but the patterns are still informative. Native birds showed a low significant tendency to move closer to cities during COVID (DiD = +0.034, $p = 0.074$), consistent with reduced human activity during lockdowns, though the result falls at the 5% threshold. Migratory birds were unaffected, since their distance barely changed during COVID, but which makes biological sense (migratory movement is driven by instinct, not local human activity). Both groups moved closer to cities during COVID, but residents did so at approximately $2.5\times$ the rate of migrants (-0.057 vs -0.022 log units), which is what the positive DiD estimate captures.

5.5 Nightlights and Bird Distance to Cities

A statistically significant negative relationship between nighttime light intensity and bird distance to city was observed across all years ($H_{1.5a}$). Within individual years, OLS regression slopes ranged from -0.0028 to -0.0071 km per radiance unit (all $p < 0.05$). Across the six-year panel, a negative NTL trend was found (slope: -0.82 radiance units/year, $r = -0.959$, $p = 0.002$), paralleled by a significant negative distance trend (slope: -5.47 km/year, $r = -0.821$, $p = 0.045$). These findings consistently indicate that as cities became relatively dimmer over the study period, birds were observed progressively farther from city centres. Native residents show $r = -0.186$ ($p < 0.001$), meaning that brighter cities are associated with shorter observation distances for resident species. Migratory birds show a weaker but still significant $r = -0.073$ ($p < 0.001$), approximately $2.5\times$ less sensitive than residents. The result is consistent with the DiD finding that residents shifted toward cities faster than migrants during COVID.

5.6 Regression with Nightlights and Pollution

Results for $H_{1.5b}$: $R^2 = 0.019$ explains $\sim 1.9\%$ of variance, the same as the nightlights model, presented in the previous section. Adding pollution does not substantially improve explanatory power, confirming that the two variables capture largely independent and individually small signals.

- `ntl_intensity` $\rightarrow$ negative coefficient (-0.0044 , $p < 0.001$): brighter cities are associated

with bird observations closer to the city centre, consistent across all specifications.

- **pollution_score_weighted** → positive coefficient (+0.013, $p < 0.001$): higher pollution is associated with observations slightly further from the city, though the effect is small and the 95% CI is wide (0.007 to 0.019), reflecting uncertainty. Ecologically plausible — heavily polluted city centres may be less suitable habitat, pushing observations outward.
- **Year** → small positive coefficient (+0.004, $p < 0.001$): a weak trend toward observations being slightly further from cities over time after controlling for NTL and pollution, likely reflecting changes in birder behaviour rather than biology.
- **month** → small positive coefficient (+0.005, $p < 0.001$): observations are slightly further from cities later in the year, consistent with seasonal movement patterns.

The correlation matrix confirms that **ntl_intensity** is the strongest correlate of distance ($r = -0.16$), while **pollution_score_weighted** shows almost no raw correlation with distance ($r = +0.02$). The pollution dataset covers only some cities, so this merged dataset is smaller and geographically biased toward well-monitored Western European cities.

Across all models, nightlight intensity is the most consistent predictor (negative and significant). Pollution adds a weak complementary signal. But, both do not explain more than $\sim 2\%$ of variance, meaning that bird observation distance is driven mainly by other factors (geography, species behaviour, and birder reporting).

In **appendix E** there are visualizations of results for sections 5.4, 5.5, 5.6.

6 Discussion

6.1 What Worked Well

We were successful in integrating the multiple heterogeneous datasets. We mapped the nightlight coordinates and extract the brightness of a city center. We combined these values with the air quality index of each city and to the GDP of that country. After extracting all the planned human expansion proxies we did feature engineer the distance of a bird sighting to the closest city center, in order to to analyse an on-going, real world phenomenon.

We were able to create an interactive folium map which shows birds observations and location with high ALAN intensity. We created a dynamic map that shows the changes of bird observations throughout the years 2017 - 2022 (files available in the repository).

6.2 Challenges and Limitations

The significant challenges we encountered were:

Observer bias in eBird: The eBird dataset reflects where people observe birds, not simply where birds are. Urban areas are over-represented because more people live there and conduct birdwatching activities nearby. Since urban areas are simultaneously brighter (high NTL) and more densely observed, using the eBird dataset creates a bias that already favours our initial assumption that high ALAN intensity lures birds into city centers. A future plan to resolve this issue is using *Zero-filling*. This method would allow us to account for where a species is reported as well as where it is *not*. Zero-filling can be done by downling the sampling event data from

eBird on top of the bird observations we had downloaded.

Small number of species per subgroup: The diurnal versus nocturnal comparison was underrepresented, with only 1 diurnal specie included. The resulting subgroup differences (hypothesis $H_{1,3i}$) should therefore not be used to make a subgroup-wide assumption.

COVID heterogeneity: The assumption of a uniform “treatment” which was needed to conduct a DiD test was violated. NTL changes during COVID varied enormously across cities. Some cities actually brightened during lockdowns (possibly due to reduced commuter traffic and increased residential lighting). This heterogeneity made causal identification difficult.

Robustness check failures for hypotheses $H_{1,3}$: The robustness checks mentioned in section 4.8 returned mostly non-significant results, undermining confidence in the original WLS findings. In our opinion, the contradicting results when the mean and median was used does not mean our results were insignificant. We hypothesize that the way we initially grouped the birds was wrong. Like mentioned in section 5.2, our results could mean that the effect is driven by a shifting subset of birds rather than a broad population-level trend. Therefore there could be other factors more specific than a birds migration period or its diurnal/nocturnal activity that makes it more sensitive to nightlights. A future plan would be to read even more about avians and group them by something more specific.

6.3 What did not work

We initially planned on incorporating a global directional migration networks dataset to directly examine trajectory deviations of some migrating bird species. However, upon further inspection, this dataset seemed to require more cleaning and feature engineering than we had expected. In the end, we did not have enough time to implement the dataset within the project timeline.

7 Conclusion

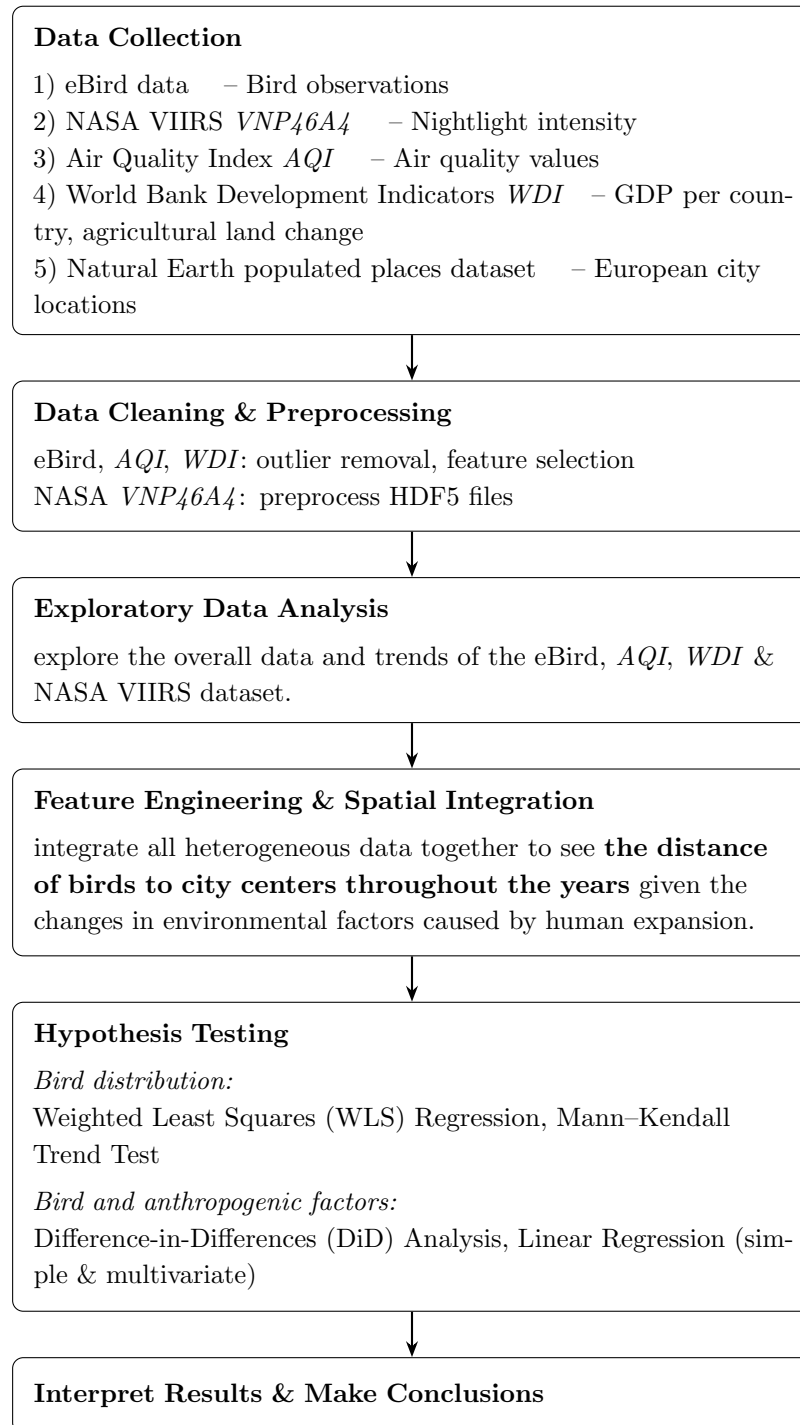
A summary of the future plans for this research include: (1) finding better ways to group birds and downloading more species for each subgroup; (2) download the sample event data from eBird to remove observer bias; (3) integrate the global directional migration networks dataset to move from point observations to route-level analysis; and (4) incorporating additional anthropogenic variables (transportation infrastructure density, land-use change indices) to build a more complete model of human footprint on avian migration. Findings from this project can add on to existing migration forecasting platforms, such as BirdCast, by providing quantitative estimates of anthropogenic adjustment factors.

This project aimed to investigate whether and to what extent anthropogenic activity influences where migratory and native birds are observed in Europe. We used artificial nighttime light intensity, air pollution, and GDP values per country as the proxies for human activity. Throughout the years 2017 to 2022, we found that nocturnal birds moved more closer to city centers than diurnal birds. Furthermore, we found a statistically significant relationship that **higher nightlight intensity was associated with birds being observed closer to urban areas**. This relationship is stronger for native than migratory species. Species type emerged as a crucial moderating variable: native birds responded more strongly to both nightlights and pollution than migratory birds.

References

- [1] C. C. M. Kyba et al., “Artificially lit surface of Earth at night increasing in radiance and extent,” *Science Advances*, vol. 3, no. 11, e1701528, 2017.
- [2] A. Fraenkel, “The growing effects of light pollution on migratory birds,” *UN Chronicle*, Oct. 6, 2022.
- [3] T. A. Van Doren et al., “High-intensity urban light installation dramatically alters nocturnal bird migration,” *Proceedings of the National Academy of Sciences*, vol. 114, no. 42, pp. 11175–11180, 2017.
- [4] U.S. Fish & Wildlife Service, “Threats to birds: Collisions (nighttime lighting),” *FWS.gov*, Dec. 6, 2022.
- [5] S. R. Loss et al., “Bird–building collisions in the United States: Estimates of annual mortality and species vulnerability,” *Biological Conservation*, vol. 212, pp. 1–9, 2017.
- [6] B. L. Sullivan et al., “eBird: a citizen-based bird observation network in the biological sciences,” *Biological Conservation*, vol. 142, pp. 2282–2292, 2009.
- [7] NASA Goddard Space Flight Center, “VIIRS Day/Night Band Nighttime Lights, Version 2,” NASA EOSDIS Land Processes DAAC, 2023. doi: 10.5067/VIIRS/VNP46A4.002.
- [8] World Bank, “World Development Indicators,” World Bank DataBank. Available at: <https://databank.worldbank.org/source/world-development-indicators>.
- [9] T. Longcore and P. A. Smith, “Artificial light at night at the global scale,” *Biological Conservation*, vol. 222, pp. 240–249, 2018.
- [10] M. Wikelski et al., “Data from space unveils a global view of animals on the move,” *Icarus – Max Planck Institute for Animal Behavior*, Mar. 20, 2022.
- [11] A. Bruederle and R. Hodler, “Nighttime lights as a proxy for human development at the local level,” *PLoS One*, vol. 13, no. 9, e0202231, 2018.
- [12] G. M. Kirwan et al., “Great Tit (*Parus major*), version 2.0,” in *Birds of the World*, Cornell Lab of Ornithology, Ithaca, NY, USA, 2024.
- [13] BirdLife International, “Species factsheet: Common Starling *Sturnus vulgaris*,” 2019. Available at: <https://datazone.birdlife.org/species/factsheet/common-starling-sturnus-vulgaris>.
- [14] B. Sun, Y. Lu, Y. Yang, M. Yu, J. Yuan, R. Yu, J. M. Bullock, N. C. Stenseth, X. Li, and J. Li, “Urbanization affects spatial variation and species similarity of bird diversity distribution,” *Science Advances*, vol. 8, no. 49, p. eade3061, 2022. <https://doi.org/10.1126/sciadv.ade3061>.
- [15] C. M. Kavwele, C. J. Torney, T. A. Morrison, S. Fulford, M. M. Masolele, J. Masoy, and J. G. C. Hopcraft, “Non-local effects of human activity on the spatial distribution of migratory wildlife in Serengeti National Park, Tanzania,” *Ecological Solutions and Evidence*, vol. 3, no. 3, p. e12159, 2022. <https://doi.org/10.1002/2688-8319.12159>.

A Pipeline



Appendix: Overview of the analysis pipeline.

The primary programming language was Python, with many key libraries used throughout, for example:

- **Data manipulation:** pandas, numpy
- **Geospatial analysis:** geopandas, shapely, pyproj
- **Raster processing:** rasterio, h5py (for VIIRS HDF5 files)

- **Statistical testing:** `scipy`, `statsmodels`, `pymannkendall`
- **Visualisation:** `matplotlib`, `seaborn`, `folium`

B Robusness Checks Results

Table 3: Summary of statistical test results for all bird subgroups. 1 = significant, 0 = not significant, NaN = test not applicable. The AND operation indicates whether all applicable tests agreed on significance.

Subgroup	Mean W.OLS	Mean MK	Median W.OLS	Median MK	AND operation
Native	1	0	0	0	Inconclusive
Migrating	1	1	0	0	Inconclusive
Diff. n.m.	1	NaN	0	NaN	Inconclusive
Spring	1	1	0	0	Inconclusive
Autumn	1	0	0	0	Inconclusive
Diff. s.a.	0	NaN	1	NaN	Inconclusive
Diurnal	0	0	1	0	Inconclusive
Nocturnal	1	1	0	1	Inconclusive
Diff. d.n.	1	NaN	1	NaN	1

C Temporal Analysis

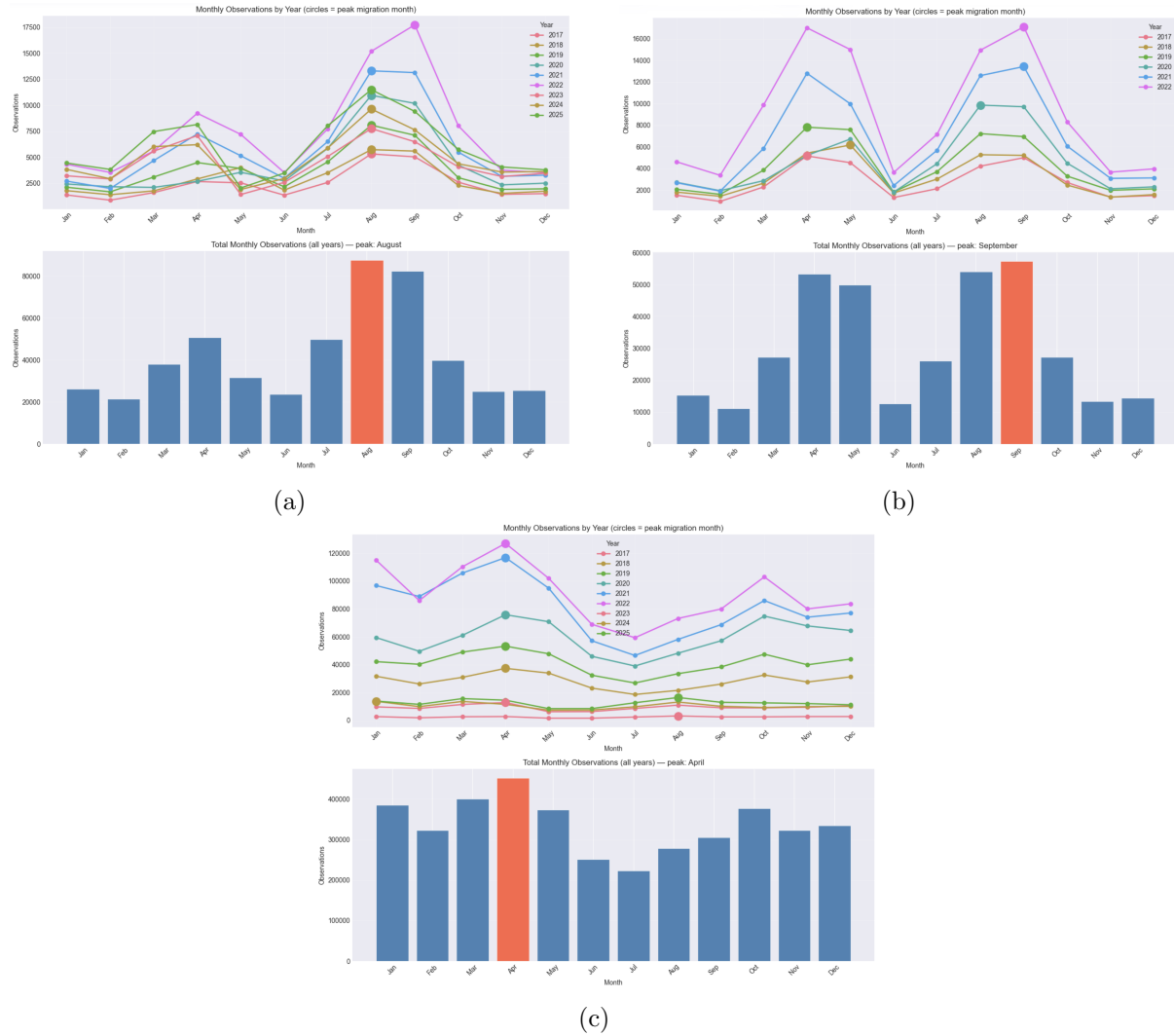
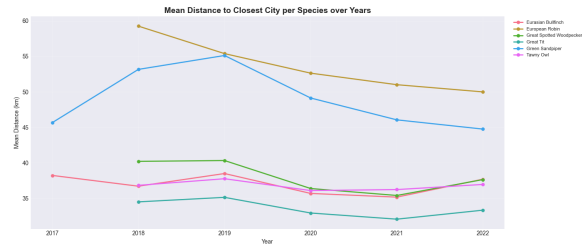
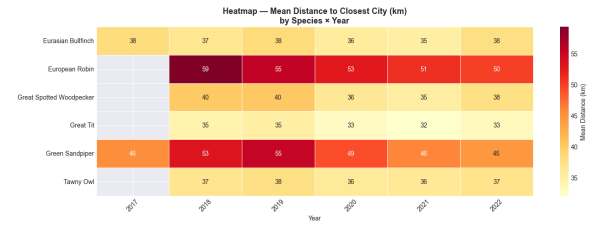


Figure 2: Temporal distribution of bird observations across Europe (2017–2025), broken down by species group. (a) Fall migrating birds, peaking in August. (b) Spring and fall migrating birds, showing two prominent peaks in spring (April–May) and autumn (September). (c) Native birds, peaking in April. In each subplot, the top panel shows monthly observation counts per year with filled circles marking the peak migration month, and the bottom panel aggregates all years to show the overall seasonal pattern.

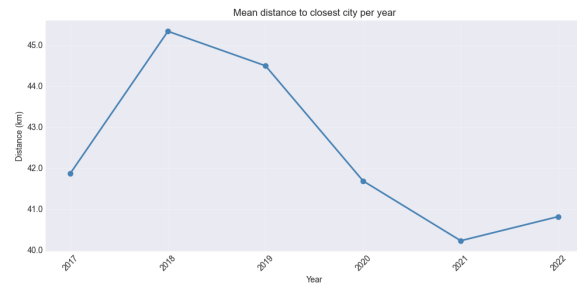
D Distance of Birds to City



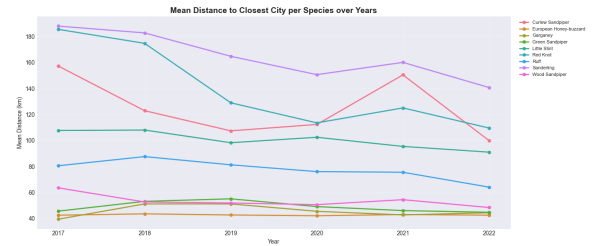
(a) Mean distance to closest city per native species over years (2017–2022).



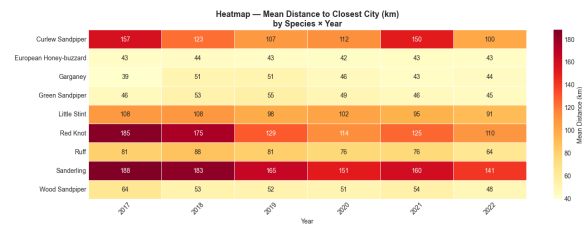
(b) Heatmap of mean distance to closest city (km) per native species and year.



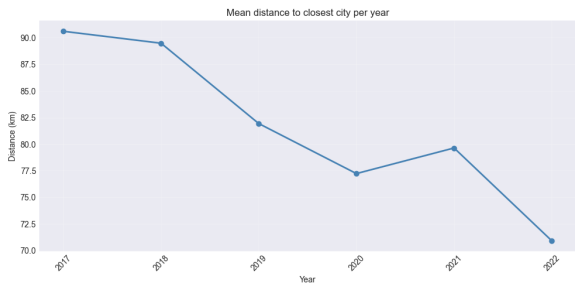
(c) Mean distance to closest city per year, aggregated across all native species (2017–2022).



(d) Mean distance to closest city per migratory species over years (2017–2022).

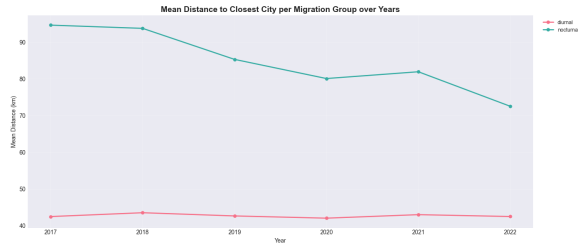


(e) Heatmap of mean distance to closest city (km) per migratory species and year.

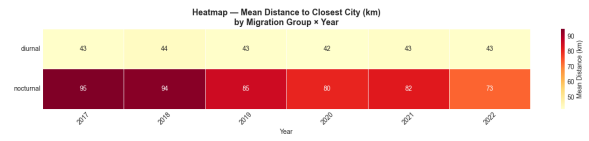


(f) Mean distance to closest city per year, aggregated across all migratory species (2017–2022).

Figure 3: Mean distance of bird observations to the closest city centre (km) over the study period, shown separately for native species (a–c) and migratory species (d–f). Line plots show species-level trends, heatmaps provide a per-species-per-year breakdown, and aggregated plots show the overall yearly trend.

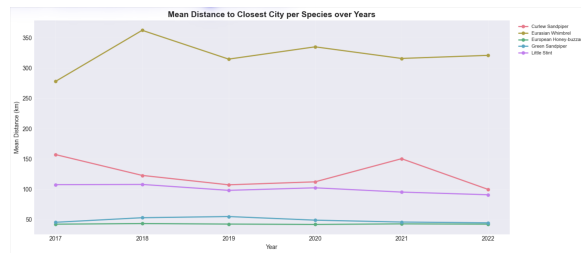


(a) Mean distance to closest city per year for diurnal and nocturnal migrating birds (2017–2022). Nocturnal birds are consistently observed closer to city centres than diurnal birds, with a clear decreasing trend over the study period.

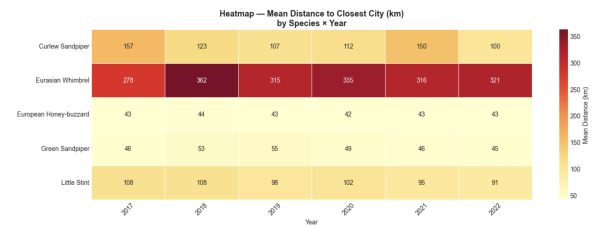


(b) Heatmap of mean distance to closest city (km) by migration group and year. Nocturnal birds (95–73 km) are observed substantially closer to cities than diurnal birds (43 km) across all years, with nocturnal distances showing a consistent decline from 2017 to 2022.

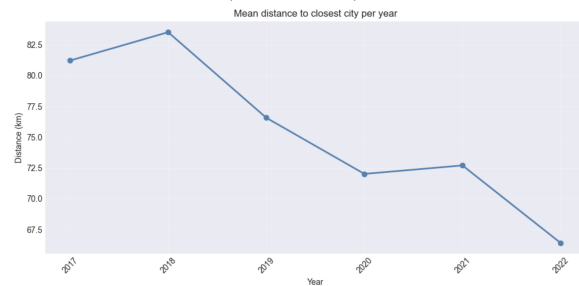
Figure 4: Comparison of mean distance to closest city centre (km) between diurnal and nocturnal migrating birds (2017–2022). Nocturnal migrants are observed consistently and substantially closer to urban areas than diurnal migrants, with nocturnal distances declining steadily over the study period.



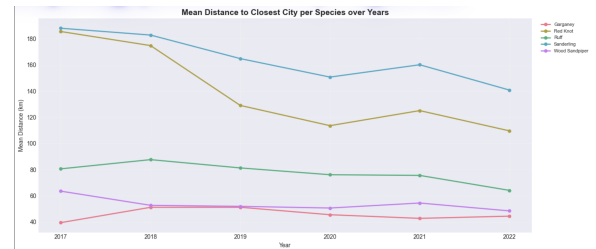
(a) Mean distance to closest city per fall migrating species over years (2017–2022).



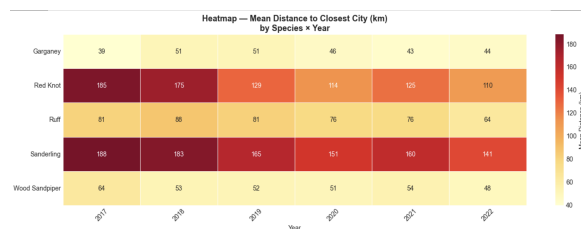
(b) Heatmap of mean distance to closest city (km) per fall migrating species and year.



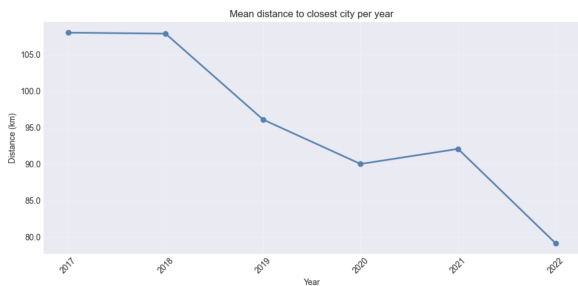
(c) Mean distance to closest city per year, aggregated across all fall migrating species (2017–2022), showing an overall decreasing trend.



(d) Mean distance to closest city per spring-and-fall migrating species over years (2017–2022).



(e) Heatmap of mean distance to closest city (km) per spring-and-fall migrating species and year.



(f) Mean distance to closest city per year, aggregated across all spring-and-fall migrating species (2017–2022), showing an overall decreasing trend.

Figure 5: Mean distance of bird observations to the closest city centre (km) over the study period, shown for fall migrating species (a–c) and spring-and-fall migrating species (d–f). Line plots show species-level trends, heatmaps provide a per-species-per-year breakdown, and aggregated plots show the overall yearly trend.

E Results Analysis

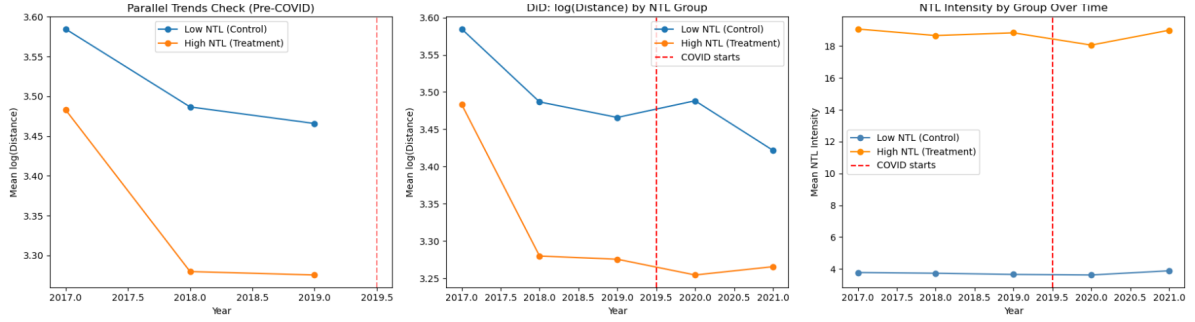


Figure 6: DiD Design 1 (H1.4a) — high vs. low NTL cities as treatment and control groups. *Left*: parallel trends check (2017–2019) showing the two groups already diverging before COVID, which violates the parallel trends assumption and limits causal interpretation. *Centre*: mean $\log(\text{Distance})$ by NTL group over the full study period, with the COVID threshold marked by the dashed red line. *Right*: mean NTL intensity by group over time, confirming the structural difference between groups. $\text{DiD} = -0.030$, $p = 0.376$ (not significant).

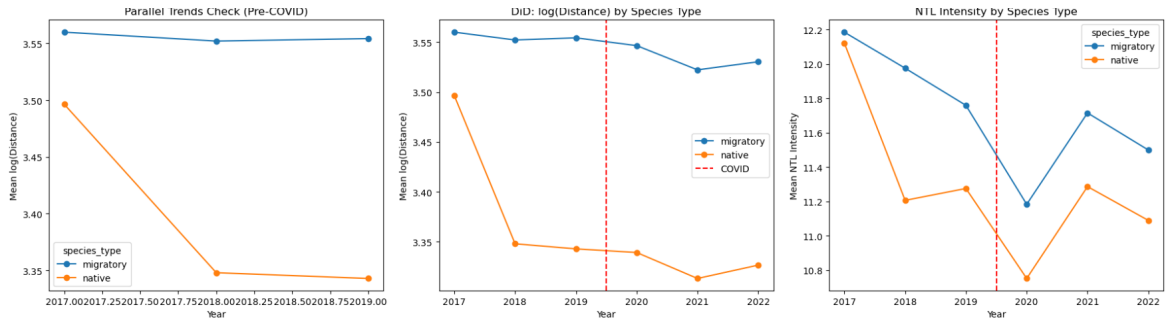
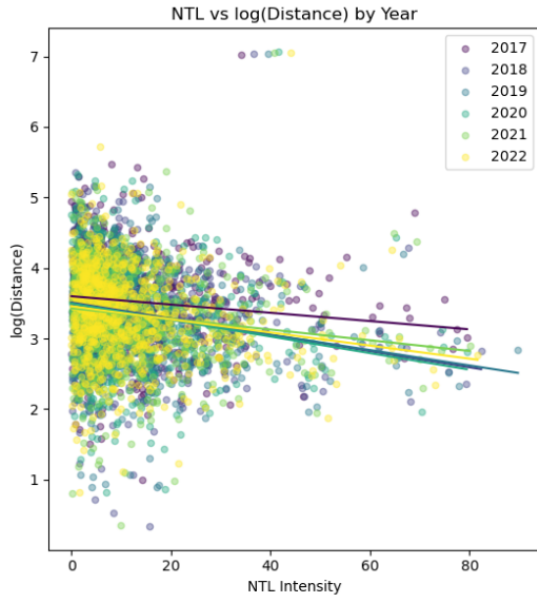
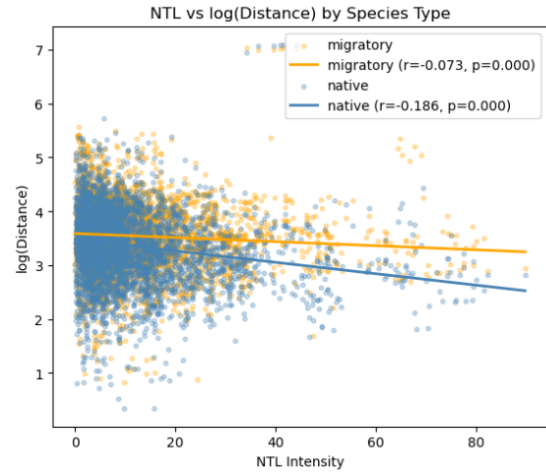


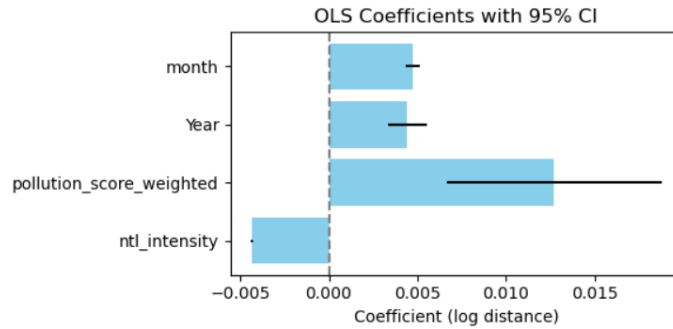
Figure 7: DiD Design 2 (H1.4b) — migratory vs. native species as treatment and control groups. *Left*: parallel trends check (2017–2019) showing native birds declining in distance while migratory birds remained stable, partially violating the parallel trends assumption. *Centre*: mean $\log(\text{Distance})$ by species type over the full study period, with the COVID threshold marked by the dashed red line. Both groups moved closer to cities during COVID, but residents did so at approximately $2.5\times$ the rate of migrants (-0.057 vs. -0.022 log units). *Right*: mean NTL intensity by species type over time. $\text{DiD} = +0.034$, $p = 0.074$.



(a) NTL intensity vs. $\log(\text{Distance})$ by year (2017–2022). Each colour represents a different year. Regression lines confirm a consistent negative slope across all years.



(b) NTL intensity vs. $\log(\text{Distance})$ by species type. Native residents show a substantially steeper negative slope ($r = -0.186$, $p < 0.001$) than migratory birds ($r = -0.073$, $p < 0.001$), indicating approximately $2.5\times$ greater sensitivity to urban light.



(c) OLS coefficient plot with 95% confidence intervals for the multivariate model including `ntl_intensity`, `pollution_score_weighted`, `Year`, and `month` ($R^2 = 0.019$). `ntl_intensity` is the only negative predictor; all others show small positive associations with observation distance.

Figure 8: NTL–distance scatter plots by year and species type, and multivariate OLS regression coefficients. The negative association between nightlight intensity and bird observation distance is present across all years and both species types, with native residents approximately $2.5\times$ more sensitive than migrants. The OLS coefficient plot confirms that nightlight intensity is the dominant and only negative predictor across all specifications, with pollution, year, and month contributing small positive effects.